CHUANRU WEI

M.S. Student in Software Engineering | Nankai University, Tianjin, China

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RESEARCH INTERESTS

Machine Learning AI Agents Game AI Procedural Content Generation Immersive Game Design Generative AI AI-native Game Creation

My research interests lie at the intersection of AI and games, particularly in exploring how learning-based and computational methods can be deeply integrated into interactive worlds. I am interested in investigating how generative and agentic AI, reinforcement learning, procedural generation, and other AI techniques can enable the generation, evolution, and interaction of immersive game worlds, spanning areas such as interactive storytelling, intelligent game characters, dynamic content generation, and human–AI co-creation.

EDUCATION

Nankai University Tianjin, China
M.S. in Software Engineering Sep. 2024 – Jun. 2027

- Average Score: **88.79/100.00**

Shandong Normal University Jinan, China
B.S. in Cyberspace Security Sep. 2020 – Jun. 2024

- Average Score: **85.84/100.00**

RESEARCH EXPERIENCE

GRAIL-VN: Graph-Grounded Runtime-Verifiable Agentic Framework for Reliable Branching Visual Novel Script Generation 2026
Co-First Author | The 22nd AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment

- Designed a persistent narrative world graph as an explicit runtime state interface between LLM-based agents and the interactive narrative environment, enabling structured tracking of characters, scenes, events, and branching conditions.
- Developed a constraint-based generation and verification pipeline integrating planning, graph alignment, structural validation, semantic judging, and iterative repair to detect and correct state and narrative inconsistencies.
- Proposed Lookahead Branch Planning and Just-In-Time Casting to improve long-horizon branch planning and dynamic character creation in open-ended narrative generation.
- Built an automated evaluation and ablation framework and evaluated 36 cross-genre and cross-complexity scenarios using metrics including structural validity, state consistency, branch-condition coverage, and goal achievement.

Drug–Target Interaction Prediction via Heterogeneous-Graph Dual-Channel Representation and Hard Negative Sampling 2025 – 2026
First Author | Journal of Computer Applications

- Designed a Similarity-Enhanced Heterogeneous Interaction Graph to strengthen drug–drug and target–target connections using feature similarity.
- Developed a Dual-Channel Encoder that combines relation-aware GraphSAGE aggregation with biased random walks and skip-gram-based semantic neighborhood modeling, jointly capturing high-order structural and cross-modal semantic information.
- Proposed a Tempered-Quota Cluster-Aware Hard Negative Sampling strategy that adaptively allocates sampling quotas across drug–target cluster blocks and performs difficulty-aware sampling within each block, improving hard-negative coverage and cross-cluster balance.

T-FDR: Trust-guided Frequency-aware Diffusion Refinement for Reliable Cardiac MRI Segmentation under Limited Annotations 2025 – 2026
Second Author | Computerized Medical Imaging and Graphics

- Contributed to the implementation and experimental evaluation of the proposed diffusion-based cardiac MRI segmentation framework.
- Participated in model development, comparative experiments, ablation studies, and analysis of segmentation performance under limited-annotation settings.

PUBLICATIONS

[1] **Wei, C.**, et al. *GRAIL-VN: A Graph-Grounded Runtime-Verifiable Agentic Framework for Reliable Branching Visual Novel Script Generation.* The 22nd AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE), 2026. Accepted as Poster. **Co-First Author**.

[2] **Wei, C.**, et al. *Drug-target interaction prediction via heterogeneous-graph dual-channel representation and hard negative sampling. Journal of Computer Applications*, 2026. Accepted for publication. **First Author**. (In Chinese).

[3] Chen, K., **Wei, C.**, et al. *T-FDR: Trust-guided frequency-aware diffusion refinement for reliable cardiac MRI segmentation under limited annotations. Computerized Medical Imaging and Graphics*, 2026. Under Revision. **Second Author**.

INDUSTRY EXPERIENCE

Baidu Beijing, China
AI / Software Engineering Intern Sep. 2025 – Dec. 2025

- Contributed to an AI-enabled AIOps platform for mobile application release quality monitoring, diagnosis, alerting, and risk mitigation.
- Designed and independently implemented core tools for an intelligent assistant, expanding its capabilities through tool-based interaction and natural-language task execution.
- Improved the assistant's intent recognition through prompt optimization and contributed a fallback RAG pipeline for unresolved user intents.
- Optimized global alerting strategies, reducing irrelevant alerts by approximately 80%, and improved high-latency tool execution through multithreaded concurrency.
- Contributed to knowledge-graph construction and tool orchestration for complex diagnostic workflows.

Meituan Beijing, China
Software Engineering Intern Apr. 2026 – Jun. 2026

- Developed CLIHub, a CLI management platform designed to provide AI agents with standardized, low-overhead access to internal RESTful services.
- Designed a unified abstraction layer for authentication and interface invocation, allowing agents to interact with backend services through standardized CLI commands.
- Designed a lightweight command discovery mechanism requiring approximately 300 tokens to retrieve interface capabilities and parameter schemas, reducing token consumption by over 90% compared with loading complete API documentation.
- Implemented structured JSON-based command outputs and a Spring Boot + Vue management console for interface registration, permission configuration, command preview, and invocation logs.

AWARDS

- National Third Prize, 12th Blue Bridge Cup National Software and Information Technology Talent Competition, 2021
- Shandong First Prize, 12th Blue Bridge Cup National Software and Information Technology Talent Competition, 2021
- North China Regional Second Prize, 2nd China Collegiate Computing Contest – AIGC Innovation Competition, 2025
- Shandong First Prize, 20th Shandong Provincial College Students' Software Design Competition, 2022

TECHNICAL SKILLS

- **AI / Machine Learning:** PyTorch, Transformers, Supervised Learning, Reinforcement Learning
- **LLM / Agents:** LLM Agents, ReAct, RAG, Prompt Engineering, Tool Use, MCP, Agentic Workflows, LangChain, LangGraph
- **Graph / Data:** Neo4j, MySQL, Redis, MongoDB, ElasticSearch
- **Programming:** Python, Java, Go, C++, TypeScript / JavaScript
- **Systems / Engineering:** Spring Boot, FastAPI, React, Vue, Docker, Linux, Git, Distributed Systems
- **Games / Interactive Systems:** Interactive Narrative, Procedural Content Generation, Game AI

LANGUAGES

English: IELTS Academic; currently preparing for retake
Chinese: Native